

Anamorphic Encryption: New Constructions and Homomorphic Realizations

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Abstract. The elegant paradigm of Anamorphic Encryption (Persiano *et al.*, Eurocrypt 2022) considers the question of establishing a private communication in a world controlled by a dictator. The challenge is to allow two users, sharing some secret anamorphic key, to exchange covert messages without the dictator noticing, even when the latter has full access to the regular secret keys. Over the last year several works considered this question and proposed constructions, novel extensions and strengthened definitions.

In this work we make progress on the study of this primitive in three main directions. First, we show that two general and well established encryption paradigms, namely hybrid encryption and the IBE-to-CCA transform, admit very simple and natural anamorphic extensions. Next, we show that anamorphism, far from being a phenomenon isolated to "basic" encryption schemes, extends also to homomorphic encryption. We show that some existing homomorphic schemes, (and most notably the fully homomorphic one by Gentry, Sahai and Waters) can be made anamorphic, while retaining their homomorphic properties both with respect to the regular and the covert message.

Finally we refine the notion of anamorphic encryption by envisioning the possibility of splitting the anamorphic key into an encryption component (that only allows to encrypt covert messages) and a decryption component. This makes possible for a receiver to set up several, independent, covert channels associated with a single covert key.

1 Introduction

Cryptography is one of the most fundamental privacy enabler of the modern era. However, as recently pointed out by Persiano *et al.* in [PPY22], this success heavily relies on two, often given for granted, assumptions: sender freedom and receiver privacy. The first postulates that senders can freely choose the message to be sent, the second assumes that the receiver's secret key remains uncompromised. While these assumptions are very natural in most circumstances they

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might be at stake in countries where law enforcement agencies can force users to surrender their decryption keys. In particular, in dictator-led countries, citizens might be allowed to send only contents approved by the regime, thus undermining the sender freedom assumption.

These problematic scenarios have been recently considered by Persiano *et al.* in [PPY22] where the novel paradigm of Anamorphic encryption has been introduced. In [PPY22] two flavors of the primitive are proposed, depending on which assumption one cannot rely on: sender anamorphic encryption considers scenarios where the sender freedom assumption does not hold, receiver anamorphic encryption addresses situations where receiver's private keys can be compromised.

In the latter case, the basic idea is that an anamorphic (public-key) encryption scheme can be deployed in two modes: regular and anamorphic. When used as regular, it works as expected for a standard public key encryption scheme. When deployed in anamorphic mode, on the other hand, key generation produces a public key apk with *two* associated secret keys: a, seemingly, regular one ask and a *double* one dk. Bob privately shares dk with Alice and publishes apk as his public key. Now, if the dictator forces Bob to reveal his secret key, Bob hands ask only, thus pretending that this is the only secret key in his possess. A key feature of anamorphic encryption is that the keypair $(\mathsf{apk}, \mathsf{ask})$ can be used to encrypt/decrypt messages like in regular mode. On the other hand, dk can be used as a symmetric key by Alice to encrypt an additional message that remains hidden even if ask is given to the dictator. This allows Alice to encrypt two messages: an innocent looking message m and a covert one $\widehat{m}$. The resulting (anamorphic) ciphertext reveals either m, when decrypted regularly (i.e. with secret key ask), or $\widehat{m}$ when decrypted anamorphically using dk. What makes this notion meaningful is the requirement that standard ciphertexts should be indistinguishable from anamorphically created ones [PPY22].

When it comes to realizing this notion, Persiano *et al.* [PPY22] argued that to effectively address privacy needs in the presence of a dictator one cannot just introduce new, more powerful schemes, as these would be immediately banned as illegal. The intriguing question is thus to show that *existing* constructions can be adapted to support the new need. For the receiver anamorphic case, which is the only one considered in this paper, they proposed two such constructions: one based on rejection sampling, that only supports very small sized messages, and a second one, based on the well known Naor-Yung transform [NY90]. This latter solution is very simple and neat. Moreover, as it is of some relevance to one of our contributions we briefly recall it here. Informally, the NY transform consists in encrypting a message m via two independent instances of a public key encryption scheme as follows

$$(\mathsf{Enc}(\mathsf{pk}_0, m), \mathsf{Enc}(\mathsf{pk}_1, m), \pi)$$

where π is a NIZK that the two ciphertexts contain the same message. In the realization from [PPY22], the anamorphic secret key is sk_0 whereas the double key is $(\mathsf{sk}_1, \mathsf{aux})$ where aux is the auxiliary information associated with the NIZK (that allows to cheat and to encrypt two different messages).

A limitation of this and related constructions however is that regular and anamorphic keys need to be generated at the same time. Once a key pair is created not in anamorphic mode it becomes impossible to associate to it a double key. In other words, it is not possible to create an anamorphic channel for a public key already in usage. This definitional limitation was recently addressed by [BGHM23] who modified the model by allowing double keys to be generated independently of key pairs. Because of this, they called *anamorphic extension* (rather than *anamorphic triplet* as in the original paper) the set of algorithms associated with the anamorphic mode. In the same paper, they proposed a notion of robustness for anamorphic encryption that, informally, aims at capturing the requirement that when (anamorphically) decrypting a regular message one should get some error message signalling that the ciphertext does not contain any covert message[1]. Banfi *et al.* [BGHM23] gave solutions achieving both these novel properties. A drawback of these solutions is that they either rely on the assumption that sender and receiver share a (synchronized) counter or that the underlying encryption scheme satisfies what they call *selective randomness recoverability* (see [BGHM23] for details).

It is thus natural to ask if these properties remain achievable even when starting from schemes that neither require synchronization nor satisfy selective randomness recoverability. As a first contribution of this paper, we give a positive answer to this question. We show that two very popular encryption mechanisms are anamorphic and allow both anamorphic extensions and robustness in a natural way. Our first construction turns any hybrid encryption schemes into an anamorphic one, whereas our second realization renders anamorphic the celebrated IBE-to-CCA transform by Boneh *et al.* [BCHK07]. Both these construction are very simple and, we stress, they make no requirement whatsoever on the underlying building blocks.

Another limitation of existing work is that, so far, existing (anamorphic) constructions only concerns rather standard encryption schemes. Yet, given the growing importance of primitives like homomorphic encryption it is (again) natural to ask if existing realizations can be proved anamorphic. We give a positive answer to this question and show that both the lifted variant of Cramer-Shoup lite [CS98] (which is linearly homomorphic and IND-CCA1 secure) and the GSW fully homomorphic encryption scheme [GSW13] can be made anamorphic. We also show that (a revisited version of) the Naor-Yung anamorphic construction from [PPY22] becomes fully homomorphic (while retaining its anamorphic properties) when replacing its basic building blocks with fully homomorphic counterparts (i.e. fully homomorphic encryption [Gen09] and fully homomorphic NIZK [ADKL19]).

As a final contribution, we further refine the notion of anamorphic encryption by envisioning the possibility of splitting the double key into a component that

[1] As argued in [BGHM23] the notion of robustness is relevant for security: a dictator could try to trick receivers to expose their possession of a double key by sending them regular (i.e. not containing any covert message) ciphertext and monitor the reaction.

allows only to encrypt and a *different* one that allows to decrypt the covert message. This modification opens the way to a novel variant of the basic primitive, that we call *fully asymmetric* and that we discuss more in detail below.

1.1 Our Contributions, More in Detail

Here we discuss each one of our contributions highlighted above separately and more in details.

Novel Examples of Anamorphism. We begin by showing that two very popular encryption mechanisms/transformations are anamorphic almost out of the box. The two mechanisms are (generic) hybrid encryption and the (MAC based) IBE-to-CCA transform from [BCHK07]. Both realizations are very simple and rely on the existence of a symmetric encryption with pseudorandom ciphertexts [Möl04,KPP+23b,KPP+23a] $\mathsf{prE.Enc}$. The basic idea is very simple (and essentially the same for both schemes). Here we discuss it for the case of hybrid encryption. Recall that hybrid encryption combines asymmetric and symmetric encryption to get the benefits of both. In a nutshell, to encrypt a message m one first chooses a random secret key k for the symmetric scheme. The message m is then symmetrically encrypted and k is encrypted using the asymmetric scheme. Turning this into an anamorphic encryption scheme only affects the way k is generated: rather than being randomly sampled, k is computed as $\mathsf{prE.Enc}(\mathsf{dk}, \widehat{m})$, where $\widehat{m}$ is the covert message and dk the double key. Notice that such a key is indistinguishable from a regular one if $\mathsf{prE.Enc}$ has pseudorandom ciphertexts. Adding robustness is also easy. The idea is to use a PRF F to embed a "secret" check when encrypting an anamorphic message. Specifically we let $k = y_1||y_2$, where $y_1 = \mathsf{prE.Enc}(\mathsf{dk}_1, \widehat{m})$, $y_2 = F_{\mathsf{dk}_2}(y_1)$ and $\mathsf{dk} = (\mathsf{dk}_1, \mathsf{dk}_2)$. The (anamorphic) decryption algorithm outputs some error message if the symmetric key does not satisfy $y_2 = F_{\mathsf{dk}_2}(y_1)$. Clearly, the usage of a PRF guarantees that, unless with very small probability, the check passes only when the ciphertext contains some covert message. Notice also that since the double key dk is totally independent from the regular key material the construction can be naturally framed in the context of anamorphic extensions.

Anamorphic Encryption with Homomorphic Properties. Current examples of anamorphic encryption schemes only concerns encryption schemes with no extra functionalities (i.e. beyond security guarantees). A main contribution of this paper is to show that, somewhat surprisingly, anamorphism is a property that can be established even in the context of homomorphic encryption, thus allowing for the possibility of performing the same homomorphic operations both on the regular and on the covert plaintext.

As a simple motivating example, imagine that some hospital ward maintains on a remote server (possibly controlled by the dictator) medical records for its patients. To give semblance that the dictator cares for citizens privacy, the latter requires the records to be (homomorphically) encrypted, so that privacy

preserving computations on these data can be done. The dictator might however impose strong constraints on (some of) these data (say, those regarding side effects of patients that got vaccinated using the vaccines produced by dictator-related companies). Using HAE one could use the anamorphic component to *also* encrypt the real data and be able to perform reliable computations on them. Notice that it seems crucial here that Eval works exactly in the same way both when operating on normal ciphertexts and on anamorphic ones. Indeed, in this way the server need not to know whether the ciphertexts it is working on are anamorphic or not. More in general, given the revolutionary impact that the concept of (fully) homomorphic encryption had in cryptography, we believe that investigating the anamorphic nature of existing homomorphic constructions is a relevant research direction that could lead to interesting and unexpected applications.

As mentioned above, as a first warm up result in this sense, we show that a revisited version of the Naor-Yung instantiation from [PPY22], becomes fully homomorphic when replacing the basic building blocks (i.e. IND-CPA secure encryption and NIZK) with fully homomorphic counterparts [Gen09, ADKL19].

In this technical overview, we discuss more in detail the main ideas underlying our, more interesting and practically relevant, Cramer Shoup lite and GSW based solutions. As per the first solution recall that a (lifted) CS-lite ciphertext is of the form

$$u_1 = g_1^r, \quad u_2 = g_2^r, \quad e = h^r g_1^m, \quad v = c^r$$

where $\mathsf{pk} = (g_1, g_2, h, c)$, m is a (small) message and $\mathsf{sk} = (x_1, x_2, z)$ is such that $c = g_1^{x_1} g_2^{x_2}$ and $h = g_1^z$. A first idea, that does not really do the job, is that, under DDH, the ciphertext above is indistinguishable from

$$u_1 = g_1^r, \quad u_2 = g_2^r g_1^{\widehat{m}}, \quad e = h^r g_1^m, \quad v = c^r$$

Thus one could use the u_2 component as a covert channel for the (small) anamorphic message $\widehat{m}$. The trouble with this idea is that a dictator in possession of sk can easily tell apart anamorphic ciphertexts from regular ones by just checking if $v^r = u_1^{x_1} u_2^{x_2}$. Our final solution (see Sect. 4.2 for the complete details) overcomes this difficulty by setting the anamorphic ciphertext as

$$u_1 = g_1^r, \quad u_2 = g_2^r g_1^{\widehat{m}}, \quad e = h^r g_1^m, \quad v = c^r g_1^{\widehat{m} x_2}$$

(which now passes the verification test) and by setting the double key dk so to allow to do this without explicitly revealing x_2.

Our GSW-based construction [GSW13] is a bit more involved. Informally, in GSW, to encrypt a message μ one produces a ciphertext C, which is an $n \times n$ matrix (with small entries) of the form[2] $\mu I_n + RA$, where A is in the public key and R is a random binary matrix. The secret key is an (approximate) eigenvector $\mathbf{v}$, for C, i.e. $\mathbf{v}$ is such that $C\mathbf{v} = \mu\mathbf{v} + \mathbf{e}$ where $\mathbf{e}$ is a small norm noise vector. Thus, the encryption of μ is a matrix C such that the secret key is an

[2] To better illustrate our basic ideas, we ignore the flattening step [GSW13] here.

(approximate) eigenvector of C with corresponding eigenvalue μ. To render this construction anamorphic the idea is to modify the public parameter generation so that ciphertexts can be created with respect to *two* secret approximate eigenvectors $\mathbf{v}_1, \mathbf{v}_2$ so that $C\mathbf{v}_1 = \mu_1\mathbf{v}_1 + \mathbf{e}_1$ with μ_1 being the "regular" message, whereas $C\mathbf{v}_2 = \mu_2\mathbf{v}_2 + \mathbf{e}_2$ with μ_2 being the "anamorphic" one.

To make this mechanism work, anamorphic ciphertexts are created as (again ignoring flattening)

$$\mu_1 P_1 + \mu_2 P_2 + RA$$

where P_i are matrices such that $P_i\mathbf{v}_j = 0$ if $i \neq j$ and $P_i\mathbf{v}_i = \mathbf{v}_i$. As we illustrate in Sect. 4.3 building such matrices is easy (in any, not necessarily prime, modulus q) and it can be done without knowing $\mathbf{v}_2$. Moreover, the modified scheme extends the nice homomorphic properties of the original scheme both to μ_1 and to μ_2.

An interesting feature of both our CS-lite and GSW-based solutions is that their anamorphism can be proved via a *tight* reduction to the *same* assumption used to prove secure the corresponding regular schemes (namely DDH and LWE respectively).

Refining the Notion. An interesting feature of the CS-lite construction highlighted above is that it allows to create a double key dk that behaves like an asymmetric encryption key. Specifically, it allows to encrypt anamorphic messages but not to decrypt anamorphic ciphertexts. In particular, such a dk does not act as a symmetric covert key[3] as imagined in the original definition from [PPY22]. This observation opens the way to a very natural generalization of anamorphic encryption where the receiver is allowed to keep a, possibly empty, secret value tk (not shared with anyone). Such a tk is not part of the regular secret key but rather the secret decryption counterpart of dk. This simple change allows a more fine grained partitioning of the (anamorphic) secret key: ask is the part that syntactically matches the real secret key (i.e. the part that one might be forced to hand to the dictator); dk is the portion of the key shared with the sender that allows the latter to create anamorphic ciphertexts. Finally, tk is an additional key that the receiver keeps secret both from the dictator and the sender and that, together with dk allows to decrypt anamorphic ciphertexts. Clearly, when setting tk as the empty string one goes back to the original definition. With this change in mind we introduce the notion of *fully asymmetric* anamorphic encryption, which, informally guarantees the IND-CPA security of plaintexts (both regular and anamorphic) even with respect to users owning the double key dk. This notion is reminiscent to that of Single Receiver anamorphic encryption (SRAE) from [KPP+23b]. What makes our notion stronger, is the fact that SRAE, when dealing with users owning dk, only guarantees the privacy of regular messages. A more precise relation between the two notions is discussed in the full version of the paper.

[3] We remark here that our NY-based construction achieves this nice property as well.

Bandwidth Rate of Our Constructions. In [PPY22] the bandwidth rate for anamorphic encryption was defined as the number of anamorphic bits transmitted divided by number of regular bits. As a final note, we remark that, all our constructions have fairly high bandwidth rates. Specifically, our homomorphic constructions all achieve bandwidth rate 1. That is, each ciphertext cointaining ℓ plaintext bits also carries ℓ covert bits. Our hybrid encryption (and IBE to CCA) based constructions, on the other hand, achieve bandwidth rate of k/n (and $(k/2)/n$ when considering their robust variants), where n is the length of regular messages and k the key size for the underlying encryption scheme with pseudorandom ciphertexts. In all cases, as required in [PPY22], covert communication can start with *zero latency* (i.e. the anamorphic system can be ready to use whenever the regular one is).

1.2 Other Related Work

The notion of anamorphic encryption is similar to several other notions, such as key-escrow (e.g. [Mic93, Bla94, FY95]), deniable encryption (e.g. [CDNO97]), kleptography (e.g. [YY96, YY97]) and public key steganography (e.g. [vH04]), but it is different in various aspects. We refer to the work of Persiano et al. [PPY22] for an in-depth comparison with these notions.

In [KPP+23b] further refinements of the notion of Anamorphic Encryption are proposed. In a nutshell, they distinguish the notions of *Multiple Receiver* and *Single Receiver* anamorphic encryption depending on wether the holder of dk has access to the regular messages like the dictator or not.

In [KPP+23a] the authors consider an even more extreme scenario where all communications must pass via a central authority (controlled by the dictator) that makes the usage of encryption even more problematic. They suggest the notion of anamorphic signature as a way to way to send covert messages via the authentication channels provided by signatures. More precise details can be found in [KPP+23a].

Concurrent and Independent Work. In [WCHY23] Wang *et al.* have introduced the notion of *strongly* secure ℓ-sender anamorphic encryption. In this notion, it is required that the security of a sender AE scheme also holds when the adversary has access to all the public and secret keys except for the secret key of the "real" receiver. The requirement is essentially the same as in our definition of (receiver) *Asymmetric* AE that can be found in the full version of the paper, but in the sender AE context. Indeed, they also show how to construct a receiver AE from a strongly secure ℓ-sender AE, which match exactly the definition of Asymmetric AE.

2 Preliminaries

2.1 Notation

We denote with $\lambda \in \mathbb{N}$ a security parameter. By PPT algorithm $\mathcal{A}$ we mean a randomized algorithm for which there exists a polynomial $p(\cdot)$ that for every

input x the running time of $\mathcal{A}(x)$ is bounded by $p(|x|)$. A function $f : \mathbb{N} \to \mathbb{R}^+$ is called *negligible* if for every positive polynomial $p(\cdot)$ there exists a $\lambda_0 \in \mathbb{N}$ such that, for every $\lambda > \lambda_0$ it holds that $f(\lambda) < 1/p(\lambda)$. We use $\mathsf{negl}(\lambda)$ to denote a generic negligible function.

Let S be a set, we denote by $x \xleftarrow{\$} S$ the uniform and random sampling of an element x from the set S. Let $\mathcal{A}$ be a probabilistic algorithm, $y \xleftarrow{\$} \mathcal{A}(\cdot)$ denotes the process of running $\mathcal{A}$ and assign the result to y. To represent the empty string we use ϵ. With $x \| y$ we denote the concatenation of strings x and y.

We denote with $\stackrel{p}{=}$ the perfect indistinguishability between two distribution ensembles, i.e., the two distributions are exactly the same. With $\stackrel{s}{=}, \stackrel{c}{=}$ we denote respectively statistical and computational indistinguishability, i.e., the two distribution ensembles appear the same to any unbounded (resp. PPT) algorithm. In case of a tuple we use a dot notation to refer to specific entries, e.g. given $\mathsf{ct} = (u_1, u_2, e, v)$ we write $\mathsf{ct}.e$ to refer to the element e in the tuple ct.

Formal descriptions of well known hardness assumptions (i.e. DDH and LWE) are deferred to the full version of the paper.

2.2 Symmetric Encryption with Pseudorandom Ciphertexts

We recall the definition of a symmetric encryption scheme with pseudorandom ciphertext from [Möl04, KPP+23b, KPP+23a]. Let n and l be polynomially bounded and $\mathsf{prE} = (\mathsf{KGen}, \mathsf{Enc}, \mathsf{Dec})$ be a symmetric encryption scheme that encrypts $n(\lambda)$-bit plaintexts into $l(\lambda)$-bit ciphertexts. We define the game $\mathsf{PRCtG}^b_{\mathsf{prE},\mathcal{D}}(\lambda)$, for $b \in \{0, 1\}$, as:

$\mathsf{PRCtG}^b_{\mathsf{prE},\mathcal{D}}(\lambda)$
$K \xleftarrow{\$} \mathsf{prE.KGen}(\lambda)$
return $\mathcal{D}^{\mathcal{O}pr^b(K,\cdot)}()$ where
$\mathcal{O}pr^0(K, m)$ returns a random $l(\lambda)$-bit string
$\mathcal{O}pr^1(K, m) = \mathsf{prE.Enc}(K, m)$

We define the advantage of an adversary $\mathcal{D}$ in distinguish between $\mathsf{PRCtG}^0_{\mathsf{prE},\mathcal{D}}(\lambda)$ and $\mathsf{PRCtG}^1_{\mathsf{prE},\mathcal{D}}(\lambda)$ as

$$\mathsf{Adv}^{\mathsf{PRCtG}}_{\mathcal{D},\mathsf{prE}}(\lambda) = \left|\Pr\left[\mathsf{PRCtG}^0_{\mathsf{prE},\mathcal{D}}(\lambda) = 1\right] - \Pr\left[\mathsf{PRCtG}^1_{\mathsf{prE},\mathcal{D}}(\lambda) = 1\right]\right|$$

Definition 1. *Let* $\mathsf{prE} = (\mathsf{KGen}, \mathsf{Enc}, \mathsf{Dec})$ *be an* IND-CPA *secure symmetric encryption scheme.* prE *has pseudorandom ciphertexts if for every* PPT *adversary* $\mathcal{D}$ *we have*

$$\mathsf{Adv}^{\mathsf{PRCtG}}_{\mathcal{D},\mathsf{prE}}(\lambda) \leq \mathsf{negl}(\lambda)$$

2.3 Homomorphic Encryption

Informally, an Homomorphic Encryption scheme is an encryption mechanism that allows to perform computations on encrypted data without having to decrypt the data first. The output of the resulting computation remains in encrypted form and, once decrypted, it coincides to what one would have been obtained performing the computation on the original plaintexts. Here we recall some basic definitions related to this primitive.

Definition 2 (Partially Homomorphic Encryption). *Let $\mathcal{F} = \cup\mathcal{F}_\ell$, for $\ell \in \mathbb{N}$, be a class of functions where every $f \in \mathcal{F}_\ell$ maps $\mathcal{M}^\ell$ to $\mathcal{M}$. An $\mathcal{F}$-homomorphic PKE scheme is an* IND-CPA *secure PKE scheme* $(\mathsf{KGen}, \mathsf{Enc}, \mathsf{Dec})$ *with message space $\mathcal{M}$ and public key space $\mathcal{PK}$ such that there exists a* PPT *algorithm* $\mathsf{Eval} : \mathcal{PK} \times \mathcal{F}_\ell \times \mathcal{C}^\ell \to \mathcal{C}$ *such that for every* $(\mathsf{pk}, \mathsf{sk}) \xleftarrow{\$} \mathsf{KGen}(\lambda), \ell = poly(\lambda), m_1, \ldots, m_\ell \in \mathcal{M}$ *and $f \in \mathcal{F}_\ell$ of description size at most $poly(\ell)$ it holds that:*

- $\mathsf{ct} \xleftarrow{\$} \mathsf{Eval}(\mathsf{pk}, f, \mathsf{Enc}(\mathsf{pk}, m_1), \ldots, \mathsf{Enc}(\mathsf{pk}, m_\ell))$ *has length at most* $\mathsf{poly}(\lambda)$.
- $\mathsf{Dec}(\mathsf{sk}, \mathsf{ct}) = f(m_1, \ldots, m_\ell)$.

Definition 3 (Fully Homomorpic Encryption). *A partially homomorphic scheme defined on the set of all functions $\mathcal{F}$, where the description of a function is a circuit, is a Fully Homomorpic PKE scheme.*

The notion of strong homomorphism, informally, requires that the ciphertexts produced by the Eval algorithm are distributed as freshly generated ones. Formally, let us consider the following distribution ensembles.

$$\mathsf{Fresh}_{f,m}(\lambda) = \{(\mathsf{pk}, c, c') : (\mathsf{sk}, \mathsf{pk}) \xleftarrow{\$} \mathsf{KGen}(\lambda), \\ c \xleftarrow{\$} \mathsf{Enc}(\mathsf{pk}, m), c' \xleftarrow{\$} \mathsf{Enc}(\mathsf{pk}, f(m))\}$$

$$\mathsf{Eval}_{f,m}(\lambda) = \{(\mathsf{pk}, c, c') : (\mathsf{sk}, \mathsf{pk}) \xleftarrow{\$} \mathsf{KGen}(\lambda), \\ c \xleftarrow{\$} \mathsf{Enc}(\mathsf{pk}, m), c' \xleftarrow{\$} \mathsf{Eval}(\mathsf{pk}, f, c)\}$$

Definition 4 (Strong Homorphism). *An $\mathcal{F}$-homomorphic PKE scheme* $(\mathsf{KGen}, \mathsf{Enc}, \mathsf{Dec}, \mathsf{Eval})$ *is said to be strongly homomorphic for a class of function $\mathcal{F}$ if, for all $\ell \in \mathbb{N}$, every $f \in \mathcal{F}_\ell$, and every input $m \in \mathcal{M}^\ell$, then holds that* $\mathsf{Fresh}_{f,m}(\lambda) \stackrel{s}{=} \mathsf{Eval}_{f,m}(\lambda)$.

The previous definition can be modified in order to obtain what is called "Perfect Strong Homomorphism" requiring that the indistinguishability between the two distribution ensembles is perfect, i.e., the two distributions are exactly the same.

A simple, yet relevant, class of homomorphic schemes is that of Linearly Homomorphic Encryption schemes. Roughly speaking, in these schemesEval allows to perform linear operations on plaintexts. In other words, the class of functions $\mathcal{F}_{lin}$ for which these schemes are designed is the class of linear functions. For clarity, in what follows we will split Eval in two subroutines: EvalScal and EvalSum.

Definition 5 (Linearly Homomorphic Encryption). *A linearly homomorphic PKE scheme, with plaintext space a group* $(\mathcal{M}, +)$ *and ciphertext space* $\mathcal{C}$*, is an* IND-CPA *secure PKE scheme* $(\mathsf{KGen}, \mathsf{Enc}, \mathsf{Dec})$ *equipped with two additional (efficient) algorithms* $\mathsf{EvalScal}$ *and* $\mathsf{EvalSum}$ *such that, for every* $(\mathsf{pk}, \mathsf{sk}) \xleftarrow{\$} \mathsf{KGen}(\lambda), \ell = poly(\lambda), m_1, m_2 \in \mathcal{M}$ *it holds that:*

- $\mathsf{EvalScal}(\mathsf{pk}, \mathsf{Enc}(\mathsf{pk}, m_1), \alpha)$ *is a* PPT *algorithm that on input the public key, an encryption of a message* m_1 *and a scalar* α*, outputs a ciphertext* $c \in \mathcal{C}$ *and it holds that* $\mathsf{Dec}(\mathsf{sk}, c) = \alpha \cdot m_1$.
- $\mathsf{EvalSum}(\mathsf{pk}, \mathsf{Enc}(\mathsf{pk}, m_1), \mathsf{Enc}(\mathsf{pk}, m_2))$ *is a* PPT *algorithm that on input the public key and the encryptions of two messages* m_1 *and* m_2 *outputs a ciphertext* $c \in \mathcal{C}$ *and it holds that* $\mathsf{Dec}(\mathsf{sk}, c) = m_1 + m_2$.

2.4 Hybrid Encryption

Hybrid encryption [Sho00], in its basic form, implements the idea of using an asymmetric encryption scheme together with a symmetric one to improve the practical efficiency of the former while avoiding the inconveniences of the latter. The idea is to use the asymmetric scheme to encrypt a freshly sampled symmetric key k, that is then used to (symmetrically) encrypt a (potentially) very large message m.

More in detail, let Π^{asy} be an asymmetric encryption scheme and Π^{sym} a symmetric encryption scheme, the hybrid scheme Π^{hyb} is presented in Fig. 1:

$\mathsf{KGen}(\lambda)$

1 : $(\mathsf{sk}, \mathsf{pk}) \xleftarrow{\$} \Pi^{\mathsf{asy}}.\mathsf{KGen}(\lambda)$
2 : **return** $(\mathsf{sk}, \mathsf{pk})$

$\mathsf{Enc}(\mathsf{pk}, m)$

1 : $k \xleftarrow{\$} \Pi^{\mathsf{sym}}.\mathsf{KGen}(\lambda)$
2 : $\mathsf{ct}_m \xleftarrow{\$} \Pi^{\mathsf{sym}}.\mathsf{Enc}(k, m)$
3 : $\mathsf{ct}_k \xleftarrow{\$} \Pi^{\mathsf{asy}}.\mathsf{Enc}(\mathsf{pk}, k)$
4 : **return** $\mathsf{ct} = (\mathsf{ct}_m, \mathsf{ct}_k)$

$\mathsf{Dec}(\mathsf{sk}, \mathsf{ct})$

1 : $k = \Pi^{\mathsf{asy}}.\mathsf{Dec}(\mathsf{sk}, \mathsf{ct}_k)$
2 : $m = \Pi^{\mathsf{sym}}.\mathsf{Dec}(k, \mathsf{ct}_m)$
3 : **return** m

Fig. 1. Hybrid Encryption Scheme Π^{hyb}.

We recall the following standard results about hybrid encryption (a similar result holds for the case of IND-CCA2 security).

Theorem 1. *[BG84] If* Π^{asy} *is a* IND-CPA *secure asymmetric encryption scheme and* Π^{sym} *is a one-time secure symmetric encryption scheme, then* Π^{hyb} *is a* IND-CPA *secure asymmetric encryption scheme.*

2.5 Anamorphic Encryption

In this section we provide the relevant definitions of Anamorphic Encryption [PPY22, KPP+23b][4]. We stress that in this paper we focus only on the notion of receiver anamorphic encryption.

Informally, an Anamorphic Encryption scheme is a PKE scheme $\Pi = (\mathsf{KGen}, \mathsf{Enc}, \mathsf{Dec})$ equipped with a so called *Anamorphic Triplet* $\Sigma = (\mathsf{aGen}, \mathsf{aEnc}, \mathsf{aDec})$ of algorithms. The scheme can be deployed in two modes: regular and anamorphic. When deployed as regular it is just like any standard PKE scheme. This means, however, that if a dictator forces Bob to reveal his secret key, the confidentiality of his communications is lost. Things change when the scheme is deployed in anamorphic mode. In such a case, Bob uses aGen to generate the anamorphic key pair $(\mathsf{apk}, \mathsf{ask})$ together with an additional *double key* dk. Bob privately shares dk with Alice and publishes apk as his public key. Now, if the dictator asks for the private key of Bob, he hands only ask, thus pretending that ask is the one and only secret key in his possess. A key feature of anamorphic encryption is that the keypair $(\mathsf{apk}, \mathsf{ask})$ can be used to encrypt messages like in normal mode, i.e., it can be used jointly with Enc and Dec resulting in a standard PKE scheme. On the other hand, dk can be used as a symmetric key by Alice to encrypt an additional message that remains hidden even when ask is given to the dictator. In particular, using aEnc with both apk and dk, Alice can encrypt two different messages: some innocent looking message m and a covert message $\widehat{m}$. The resulting (anamorphic) ciphertext act when decrypted via Dec with secret key ask reveals m. On the other hand, when decrypted with aDec with keys ask and dk reveals $\widehat{m}$.

It goes without saying that in order for this notion to be any meaningful, standard ciphertexts should be indistinguishable from anamorphic ones [PPY22].

In what follows, we give a formalization of anamorphic encryption that generalizes the notion discussed above in the following sense. We allow the receiver to keep a, possibly empty, secret value tk that is neither part of ask nor part of dk. This change allows a more fine grained partitioning of the (anamorphic) secret key: ask is the part of the secret key that syntactically matches the real secret key (i.e. the part that one might be forced to give to the dictator); dk is the portion of the secret key shared with the sender that allows the latter to create anamorphic ciphertexts (but not necessarily also to decrypt them!). Finally, tk is an additional key that the receiver keeps secret both from the dictator and the sender and that, together with dk allows to decrypt anamorphic ciphertexts. Clearly, when setting tk as the empty string one goes back to the original definition.

Definition 6 (Anamorphic Triplet). *A triplet* $\Sigma = (\mathsf{aGen}, \mathsf{aEnc}, \mathsf{aDec})$ *is said anamorphic if:*

[4] We stress that the definitions given in [KPP+23b] differs from the original definition from [PPY22] in some small details, that make the former more readily suitable to our setting. We refer the interested reader to [KPP+23b] for an in-depth discussion of these differences.

- aGen *is a* PPT *algorithm that takes as input the security parameter* λ *and outputs an anamorphic public key* apk*, an anamorphic secret key* ask*, a (possibly empty) trapdoor key* tk *and a double key* dk*.*
- aEnc *is a* PPT *algorithm that takes as input* $\mathsf{apk}, \mathsf{dk}$ *and two messages* $m \in \mathcal{M}$ *and* $\widehat{m} \in \widehat{\mathcal{M}}$*. It outputs an anamorphic ciphertext* act *for* Π*.*
- aDec *is a deterministic algorithm that on input* $\mathsf{dk}, \mathsf{tk}, \mathsf{ask}$ *and an anamorphic ciphertext* act *outputted by* aEnc *outputs the anamorphic message* $\widehat{m} \in \widehat{\mathcal{M}}$ *or the special symbol* $\perp \notin \widehat{\mathcal{M}}$*.*

We next define two games $\mathsf{RealG}_{\Pi}(\lambda, \mathcal{D})$ and $\mathsf{AnamorphicG}_{\Sigma}(\lambda, \mathcal{D})$ that we will use to define Anamorphic Encryption.

$\mathsf{RealG}_{\Pi}(\lambda, \mathcal{D})$
$(\mathsf{pk}, \mathsf{sk}) \xleftarrow{\$} \mathsf{KGen}(\lambda)$
return $\mathcal{D}^{\mathcal{O}e(\mathsf{pk},\cdot,\cdot)}(\mathsf{pk}, \mathsf{sk})$ where $\mathcal{O}e(\mathsf{pk}, m, \widehat{m}) = \mathsf{Enc}(\mathsf{pk}, m)$

$\mathsf{AnamorphicG}_{\Sigma}(\lambda, \mathcal{D})$
$((\mathsf{apk}, \mathsf{ask}), \mathsf{tk}, \mathsf{dk}) \xleftarrow{\$} \mathsf{aGen}(\lambda)$
return $\mathcal{D}^{\mathcal{O}a(\mathsf{apk},\mathsf{dk},\cdot,\cdot)}(\mathsf{apk}, \mathsf{ask})$ where $\mathcal{O}a(\mathsf{apk}, \mathsf{dk}, m, \widehat{m}) = \mathsf{aEnc}(\mathsf{apk}, \mathsf{dk}, m, \widehat{m})$

We define the advantage of an adversary $\mathcal{D}$ in distinguishing between the two games as

$$\mathsf{Adv}^{\mathsf{Anamorphism}}_{\mathcal{D},\Pi,\Sigma}(\lambda) = |\Pr[\mathsf{RealG}_{\Pi}(\lambda, \mathcal{D}) = 1] - \Pr[\mathsf{AnamorphicG}_{\Sigma}(\lambda, \mathcal{D}) = 1]|$$

Definition 7 (Anamorphic Encryption). *A PKE encryption scheme* $\Pi = (\mathsf{KGen}, \mathsf{Enc}, \mathsf{Dec})$ *is an Anamorphic Encryption scheme if it is* IND-CPA *secure and there exists an anamorphic triplet* Σ *such that for every* PPT *dictator* $\mathcal{D}$ *there exists a negligible function* $\mathsf{negl}(\lambda)$ *such that*

$$\mathsf{Adv}^{\mathsf{Anamorphism}}_{\mathcal{D},\Pi,\Sigma}(\lambda) \leq \mathsf{negl}(\lambda)$$

Remark 1. As pointed out in [PPY22], it is possible to define a notion of indistinguishability of anamorphic messages, that extends the standard notion of IND-CPA security to the case of anamorphic messages. However, as showed in [PPY22], this property follows from the fact that anamorphic ciphertexts are indistinguishable from real ones and the underlying scheme is IND-CPA secure.

As noticed in [BGHM23], in practice it might be useful to be able to split the key generation phase in two separate steps so to be able to separate the process of generating a normal key from that of generating a corresponding anamorphic one. A good motivation for wanting this flexibility is, for example, that one might want to create an anamorphic variant of some public key encryption scheme already in usage. Yet, separating the two steps when using the original definition

is problematic as the anamorphic key generation outputs a key pair and the double key. Building on the requirement that the anamorphic key pair has to be indistinguishable from a regular key pair, [BGHM23] strengthen the definition by requiring that the anamorphic key generation only outputs a double key, on input the public key of a valid key-pair. Since there are no anamorphic key pairs in the definition from [BGHM23] they introduce instead the notion of anamorphic extension. Here we give the definition of Anamorphic Extension, adapted to our syntax and to the definition of anamorphic encryption given above.

Definition 8 (Anamorphic Extension). *Let* $\Pi = (\mathsf{KGen}, \mathsf{Enc}, \mathsf{Dec})$ *be a PKE scheme with implicit public parameters* pp*, an anamorphic extension of* Π *is a triplet* $\Sigma = (\mathsf{aGen}, \mathsf{aEnc}, \mathsf{aDec})$*, where:*

- aGen *is a* PPT *algorithm that on input a public key* pk *for* Π*, outputs a (possibly empty) trapdoor key* tk *and a double key* dk*.*
- aEnc *is a* PPT *algorithm that on input a double key* dk*, a message* $m \in \mathcal{M}$ *for* Π*, and a covert message* $\widehat{m} \in \widehat{\mathcal{M}}$*, outputs an anamorphic ciphertext* $\mathsf{act} \xleftarrow{\$} \mathsf{aEnc}(\mathsf{dk}, m, \widehat{m})$ *for* Π*.*
- aDec *is a deterministic algorithm that on input a double key* dk*,* tk *and the secret key* ask *and an anamorphic ciphertext* act *for* Π*, outputs a covert message* $\widehat{m} = \mathsf{aDec}(\mathsf{dk}, \mathsf{tk}, \mathsf{ask}, \mathsf{act}) \in \widehat{\mathcal{M}}$ *or the special symbol* $\perp \notin \widehat{\mathcal{M}}$*.*

The security property for Anamorphic Extension is defined analogously to the one for Anamorphic Triplet.

Remark 2. The notion of anamorphic extension implicitly assumes that the underlying PKE can be turned into an anamorphic one. This is indeed the case for some constructions (e.g. see examples in [BGHM23] for instance). However there are also cases where this conversion does not seem to be possible. Notable examples are the homomorphic schemes discussed in Sect. 4.

Robustness of Anamorphic Encryption. Robustness for anamorphic encryption has been introduced in [BGHM23]. Informally, it requires that it should be difficult to find a message m that, when encrypted normally and then *anamorphically* decrypted (i.e. using aDec) results in some $\widehat{m} \neq \perp$.

Formally, let Π be a PKE scheme equipped with an Anamorphic Triplet Σ. We define the following game, for $b \in \{0, 1\}$.

$\mathsf{Robust}^{b}_{\Pi,\Sigma}(\mathcal{A})$

$((\mathsf{apk}, \mathsf{ask}), \mathsf{dk}, \mathsf{tk}) \xleftarrow{\$} \mathsf{aGen}(\lambda)$

return $\mathcal{A}^{\mathcal{O}^b(\mathsf{apk},\mathsf{ask},\mathsf{dk},\mathsf{tk},\cdot)}(\mathsf{apk}, \mathsf{ask})$ where

$\mathcal{O}^0(\mathsf{apk}, \mathsf{ask}, \mathsf{dk}, \mathsf{tk}, m) = \mathsf{aDec}(\mathsf{dk}, \mathsf{tk}, \mathsf{ask}, \mathsf{Enc}(\mathsf{apk}, m))$

$\mathcal{O}^1(\mathsf{apk}, \mathsf{ask}, \mathsf{dk}, \mathsf{tk}, m) = \perp$

And we define the advantage of an adversary $\mathcal{A}$ in breaking the robustness property as

$$\mathsf{Adv}^{\mathsf{rob}}_{\mathcal{A},\Pi,\Sigma}(\lambda) = \left|\Pr\left[\mathsf{Robust}^0_{\Pi,\Sigma}(\mathcal{A}) = 1\right] - \Pr\left[\mathsf{Robust}^1_{\Pi,\Sigma}(\mathcal{A}) = 1\right]\right|$$

Definition 9 (Robustness). *An Anamorphic Encryption scheme* Π *equppied with Anamorphic Triplet* Σ *is said to be robust if for all* PPT *adversary* $\mathcal{A}$ *it holds that*

$$\mathsf{Adv}^{\mathsf{rob}}_{\mathcal{A},\Pi,\Sigma}(\lambda) \leq \mathsf{negl}(\lambda)$$

Even though we presented the notion of robustness for the case of Anamorphic Encryption schemes equipped with an Anamorphic Triplet, an analogous definition can be formulated for the case of Anamorphic Encryption schemes equipped with Anamorphic Extension.

2.6 Fully Asymmetric Anamorphic Encryption

Our choice of adding the component tk to the anamorphic secret key opens the way to a novel notion of anamorphic encryption, that we call *Fully Asymmetric*. Informally, this guarantees the privacy of both the regular and the anamorphic messages with respect to users having access *also* to dk (but not to ask and tk of course). We formalize the notion of Fully Asymmetric Anamorphic triplet by means of the following game where $\mathcal{A}$ is a PPT adversary, $b \in \{0,1\}$ and $\Sigma = (\mathsf{aGen}, \mathsf{aEnc}, \mathsf{aDec})$ is an Anamorphic Triplet.

$\mathsf{FAsyAnam\text{-}IND\text{-}CPA}^b_{\Sigma}(\mathcal{A})$
$(\mathsf{apk}, \mathsf{ask}, \mathsf{dk}, \mathsf{tk}) \xleftarrow{\$} \mathsf{aGen}(\lambda)$
$(m_0, m_1, \widehat{m}_0, \widehat{m}_1) \xleftarrow{\$} \mathcal{A}(\mathsf{apk}, \mathsf{dk})$
$\mathsf{act} \xleftarrow{\$} \mathsf{aEnc}(\mathsf{apk}, \mathsf{dk}, m_b, \widehat{m}_b)$
return $\mathcal{A}(\mathsf{act})$

We define the advantage of an adversary $\mathcal{A}$ in breaking the Fully Asymmetric property as

$$\begin{aligned}\mathsf{Adv}^{\mathsf{FAsy\text{-}Anam}}_{\mathcal{A},\Sigma}(\lambda) = \Big|&\Pr\left[\mathsf{FAsyAnam\text{-}IND\text{-}CPA}^0_{\Sigma}(\mathcal{A}) = 1\right] \\ &- \Pr\left[\mathsf{FAsyAnam\text{-}IND\text{-}CPA}^1_{\Sigma}(\mathcal{A}) = 1\right]\Big|\end{aligned}$$

Notice that the adversary does not receive any (additional) encryption oracle as having both apk and dk it can create both regular and anamorphic ciphertexts on its own.

Definition 10 (Fully Asymmetric Anamorphic Encryption). *An Anamorphic Encryption scheme* Π *equppied with Anamorphic Triplet* Σ *is said to be Fully Asymmetric if for every* PPT *adversary* $\mathcal{A}$ *it holds that*

$$\mathsf{Adv}^{\mathsf{FAsy\text{-}Anam}}_{\mathcal{A},\Sigma}(\lambda) \leq \mathsf{negl}(\lambda)$$

Our formalization of Fully Asymmetric AE is reminiscent of the notion of Single Receiver Anamorphic Encryption from [KPP+23b]. What makes our notion stronger, is the fact that the latter guarantees the privacy of regular messages whereas our notion protects both regular and anamorphic messages. In the full version of the paper we make this connection more precise by showing that one can obtain a fully asymmetric AE from a Single receiver AE, if, informally, the latter realizes an asymmetric scheme for covert messages.

3 Generic Constructions

In this section we present our generic constructions of anamorphic encryption. The first is realized from any hybrid encryption, while the second builds upon the well known realization [BCHK07] of chosen ciphertext secure encryption from identity based encryption. Since both schemes build on similar ideas, the latter construction is deferred to the full version of the paper.

3.1 Construction from Hybrid Encryption

In this section we show that any hybrid encryption can be turned into an anamorphic encryption scheme as long there exists a symmetric encryption scheme with pseudorandom ciphertexts. The construction is very simple, as the basic idea is to hide the anamorphic message in the symmetric encryption key used in the hybrid encryption. A remarkable feature of our scheme is that, as detailed in the next subsection, it achieves robustness essentially for free. Moreover, the scheme can be naturally stated in the framework of anamorphic extensions.

Let Π^{hyb} be a hybrid encryption scheme and prE a symmetric encryption scheme with pseudorandom ciphertext. The anamorphic extension $\Sigma^{\mathsf{hyb}} = (\mathsf{aGen}, \mathsf{aEnc}, \mathsf{aDec})$ is defined in Fig. 2.

$\mathsf{aGen}(\mathsf{pk})$

1 : $\widehat{k} \xleftarrow{\$} \mathsf{prE.KGen}(\lambda)$
2 : $\mathsf{dk} = (\mathsf{pk}, \widehat{k})$
3 : $\mathsf{tk} = \epsilon$
4 : **return** $(\mathsf{dk}, \mathsf{tk})$

$\mathsf{aEnc}(\mathsf{dk}, m, \widehat{m})$

1 : $k \xleftarrow{\$} \mathsf{prE.Enc}(\widehat{k}, \widehat{m})$
2 : $\mathsf{ct}_m \xleftarrow{\$} \Pi^{\mathsf{sym}}.\mathsf{Enc}(k, m)$
3 : $\mathsf{ct}_k \xleftarrow{\$} \Pi^{\mathsf{asy}}.\mathsf{Enc}(\mathsf{pk}, k)$
4 : **return** $\mathsf{act} = (\mathsf{ct}_m, \mathsf{ct}_k)$

$\mathsf{aDec}(\mathsf{dk}, \mathsf{tk}, \mathsf{sk}, \mathsf{act})$

1 : $k = \Pi^{\mathsf{asy}}.\mathsf{Dec}(\mathsf{sk}, \mathsf{ct}_k)$
2 : $\widehat{m} = \mathsf{prE.Dec}(\widehat{k}, k)$
3 : **return** $\widehat{m}$

Fig. 2. Anamorphic Extension Σ^{hyb}.

For simplicity in the following proof, we write $\Pi^{\mathsf{hyb}}.\mathsf{Enc}(\mathsf{pk}, m; k)$ to denote that the key k of the symmetric encryption is given explicitly as input. Note that $\mathsf{aEnc}(\mathsf{dk}, m, \widehat{m}) = \Pi^{\mathsf{hyb}}.\mathsf{Enc}(\mathsf{pk}, m; k)$ where $k \xleftarrow{\$} \mathsf{prE.Enc}(\widehat{k}, \widehat{m})$.

Lemma 1. *If there exists a symmetric encryption with pseudorandom ciphertext* prE *then any hybrid encryption scheme* Π^{hyb} *equipped with the anamorphic extension* Σ^{hyb} *defined in Fig. 2 is an Anamorphic Encryption scheme. Namely, for all* PPT *adversary* $\mathcal{D}$ *there exists an adversary* $\mathcal{A}$ *such that*

$$\mathsf{Adv}^{\mathsf{Anamorphism}}_{\mathcal{D},\Pi^{\mathsf{hyb}},\Sigma^{\mathsf{hyb}}}(\lambda) \leq \mathsf{Adv}^{\mathsf{PRCtG}}_{\mathcal{A},\mathsf{prE}}(\lambda)$$

Proof. Let $\Sigma^{\mathsf{hyb}} = (\mathsf{aGen}, \mathsf{aEnc}, \mathsf{aDec})$ be the anamorphic extension defined above. Suppose that exists an adversary $\mathcal{D}$ that distinguishes between $\mathsf{RealG}_{\Pi^{\mathsf{hyb}}}$ and $\mathsf{AnamorphicG}_{\Sigma^{\mathsf{hyb}}}$, we can construct an adversary $\mathcal{A}$ against the pseudorandomness of prE.

$\mathcal{A}^{\mathcal{O}(\cdot)}$

1 : $(\mathsf{pk}, \mathsf{sk}) \xleftarrow{\$} \mathsf{KGen}(\lambda)$
2 : Whenever $\mathcal{D}(\mathsf{pk}, \mathsf{sk})$ makes a query, $\forall i \in \{1, \ldots, q\}$ compute:
3 : $\quad r \xleftarrow{\$} \mathcal{O}(\widehat{m})$
4 : $\quad \mathsf{act} \xleftarrow{\$} \Pi^{\mathsf{hyb}}.\mathsf{Enc}(\mathsf{pk}, m; r)$
5 : $\quad$ Answer to $\mathcal{D}$ with the ciphertext act
6 : **return** $\mathcal{D}$'s output

Fig. 3. $\mathcal{A}$ reducing a distinguisher $\mathcal{D}$ for Anamorphism to PRCtG.

Precisely, $\mathcal{A}$ has access to an oracle $\mathcal{O}(\cdot)$ that can be either a procedure that returns random string s or the result of $\mathsf{prE.Enc}(K, \widehat{m})$ for a fixed randomly selected K. Let $q = poly(\lambda)$ be the number of queries made by $\mathcal{D}$. The pseudocode of $\mathcal{A}$ is given in Fig. 3. Now we can analyze $\mathcal{D}$'s view relative to the oracle that has been provided to $\mathcal{A}$. The key pair $(\mathsf{pk}, \mathsf{sk})$ is generated by KGen, just like in the two games $\mathsf{RealG}_{\Pi^{\mathsf{hyb}}}$ and $\mathsf{AnamorphicG}_{\Sigma^{\mathsf{hyb}}}$. If $\mathcal{O}$ outputs a random string when $\mathcal{A}$ makes a query, then $\mathcal{D}$ receives a ciphertext computed using a uniformly distributed random key for the symmetric encryption scheme, so just like in a normal hybrid encryption scheme. Hence we can state that $\Pr\left[\mathsf{RealG}_{\Pi^{\mathsf{hyb}}}(\lambda, \mathcal{D}) = 1\right] = \Pr\left[\mathsf{PRCtG}^0_{\mathsf{prE},\mathcal{A}}(\lambda) = 1\right]$. Otherwise, if the oracle $\mathcal{O}$ returns an encryption of $\widehat{m}$ using prE, $\mathcal{D}$ receives a ciphertext computed using an encryption of $\widehat{m}$ with the scheme prE using the key K, just like in the anamorphic encryption algorithm. So we can state that the $\Pr\left[\mathsf{AnamorphicG}_{\Sigma^{\mathsf{hyb}}}(\lambda, \mathcal{D}) = 1\right] = \Pr\left[\mathsf{PRCtG}^1_{\mathsf{prE},\mathcal{A}}(\lambda) = 1\right]$.

So we can state that the view of $\mathcal{D}$ is perfectly simulated by $\mathcal{A}$. So, if $\mathcal{D}$ breaks the anamorphism then also $\mathcal{A}$ breaks the pseudorandomness of prE, i.e., $\mathsf{Adv}^{\mathsf{Anamorphism}}_{\mathcal{D},\Pi^{\mathsf{hyb}},\Sigma^{\mathsf{hyb}}}(\lambda) \leq \mathsf{Adv}^{\mathsf{PRCtG}}_{\mathcal{A},\mathsf{prE}}(\lambda)$.

Theorem 2. *Any hybrid encryption scheme* Π^{hyb} *that is* IND-CPA *secure is an Anamorphic Encryption scheme.*

Proof. If Π^{hyb} is IND-CPA secure then there exists a one-way function [IL89] and so a symmetric encryption scheme with pseudorandom ciphertext prE can be built. From prE we can construct the anamorphic triplet Σ^{hyb} previously described, and applying the previous lemma the theorem is proved.

A Robust Variant. To make the scheme robust, the basic idea is to use a PRF to embed a "secret" check when encrypting an anamorphic message. The properties of the PRF guarantee that, unless with negligible probability, the check passes only when aEnc has been used to create the ciphertext. Details follow.

Let prE be a symmetric encryption scheme with pseudorandom ciphertexts with keyspace $\mathcal{K}_1$ that encrypts messages in $\{0,1\}^{n_1}$ producing ciphertexts in $\{0,1\}^{n/2}$, $n/2 \geq n_1$. Let F be a PRF that maps elements of $\mathcal{K}_2 \times \{0,1\}^{n/2}$ into $\{0,1\}^{n/2}$. Let Π^{hyb} be a hybrid encryption scheme. The anamorphic extension $\Sigma^{\mathsf{hyb}}_{\mathsf{rob}} = (\mathsf{aGen}, \mathsf{aEnc}, \mathsf{aDec})$ is defined in Fig. 4. The proof of the following theorem appears in the full version of the paper.

$\mathsf{aGen}(\mathsf{pk})$

1 : $\widehat{k}_1 \xleftarrow{\$} \mathcal{K}_1$
2 : $\widehat{k}_2 \xleftarrow{\$} \mathcal{K}_2$
3 : $\mathsf{dk} = (\mathsf{pk}, \widehat{k}_1, \widehat{k}_2)$
4 : $\mathsf{tk} = \epsilon$
5 : **return** $(\mathsf{dk}, \mathsf{tk})$

$\mathsf{aEnc}(\mathsf{dk}, m, \widehat{m})$

1 : $y_1 \xleftarrow{\$} \mathsf{prE.Enc}(\widehat{k}_1, \widehat{m})$
2 : $y_2 = \mathsf{F}(\widehat{k}_2, y_1)$
3 : $k = y_1 \| y_2$
4 : $\mathsf{ct}_m \xleftarrow{\$} \Pi^{\mathsf{sym}}.\mathsf{Enc}(k, m)$
5 : $\mathsf{ct}_k \xleftarrow{\$} \Pi^{\mathsf{asy}}.\mathsf{Enc}(\mathsf{pk}, k)$
6 : **return** $\mathsf{act} = (\mathsf{ct}_m, \mathsf{ct}_k)$

$\mathsf{aDec}(\mathsf{dk}, \mathsf{tk}, \mathsf{sk}, \mathsf{act})$

1 : Parse $\mathsf{act} = (\mathsf{ct}_m, \mathsf{ct}_k)$
2 : $k = \Pi^{\mathsf{asy}}.\mathsf{Dec}(\mathsf{sk}, \mathsf{ct}_k)$
3 : Parse $k = y_1 \| y_2$
4 : **if** $F(\widehat{k}_2, y_1) = y_2$ **then**
5 : $\quad \widehat{m} = \mathsf{prE.Dec}(\widehat{k}_1, y_1)$
6 : **else**
7 : $\quad \widehat{m} = \perp$
8 : **return** $\widehat{m}$

Fig. 4. Anamorphic Extension $\Sigma^{\mathsf{hyb}}_{\mathsf{rob}}$.

Theorem 3. *If* F *is a* PRF *the proposed construction is robust. In particular, for all* PPT *adversaries* $\mathcal{D}$ *we can construct an adversary* $\mathcal{A}$ *such that*

$$\mathsf{Adv}^{\mathsf{rob}}_{\mathcal{D},\Pi^{\mathsf{hyb}},\Sigma^{\mathsf{hyb}}_{\mathsf{rob}}}(\lambda) \leq \mathsf{Adv}^{\mathsf{PRF}}_{\mathsf{F},\mathcal{A}}(\lambda) + \frac{q}{2^{n/2}}$$

where $q = poly(\lambda)$ *is the number of queries made by* $\mathcal{A}$.

4 Anamorphic Encryption with Homomorphic Properties

Here we introduce and realize the notion of Anamorphic Encryption with homomorphic properties. Informally, such a primitive, that we call Homomorphic

Anamorphic Encryption (HAE for short) is an anamorphic encryption scheme that support homomorphic operations on both the regular and the anamorphic plaintexts. We will give the definition of HAE for the case of anamorphic encryption schemes with associated anamorphic triplet as this is the setting of interest for our construction. It goes without saying that the definition can be adapted straightforwardly to the case of anamorphic extensions.

Definition 11 (Homomorphic Anamorphic Encryption). *Given an anamorphic encryption scheme* Π*, with corresponding anamorphic triplet* Σ*. The scheme is said to be a Homomorphic Anamorphic encryption scheme for the class of functions* $\mathcal{F}$ *if* Π *is an* $\mathcal{F}$*-homomorphic encryption scheme and it holds that, for every* $f \in \mathcal{F}$*:*

- $\mathsf{act}' \xleftarrow{\$} \Pi.\mathsf{Eval}(\mathsf{pk}, f, \mathsf{aEnc}(\mathsf{apk}, \mathsf{dk}, m_1, \widehat{m}_1), \ldots, \mathsf{aEnc}(\mathsf{apk}, \mathsf{dk}, m_\ell, \widehat{m}_\ell))$ *has length at most* n*.*
- $\mathsf{aDec}(\mathsf{dk}, \mathsf{tk}, \mathsf{ask}, \mathsf{act}') = f(\widehat{m}_1, \ldots, \widehat{m}_\ell)$*.*
- $\mathsf{Dec}(\mathsf{ask}, \mathsf{act}') = f(m_1, \ldots, m_\ell)$*.*

The definitions of Linearly Homomorphic Encryption and Strong Homomorphism apply naturally in this context.

4.1 Naor-Yung Transform Gives Homomorphic Anamorphic Encryption

The Naor-Yung transform [NY90], when applied to an IND-CPA secure PKE scheme Π, gives an IND-CCA1 secure encryption scheme NY. If the NIZK used is also *simulation sound* then the resulting PKE scheme is IND-CCA2 secure [Sah99]. The idea is to reach the non malleability of the ciphertexts encrypting the message m under two different public keys and to prove with a NIZK proof that the two ciphertexts encrypt the same message.

In [PPY22] they give an Anamorphic Encryption scheme based on this transform by letting the message sender know the simulation trapdoor of the NIZK, in order to encrypt two different messages, i.e. the regular one and the anamorphic one, and cheating in the proof.

Let Π and Σ be respectevely the underlying PKE scheme and NIZK of NY. The anamorphic triplet aNY is the given in Fig. 5. The proof that the resulting scheme is anamorphic was given in [PPY22].

Fully Asymmetric. The construction we have given in this paper is a bit different from the one of [PPY22]. In their construction sk_1 is given in dk, and so all *anamorphic senders* can decrypt an anamorphic ciphertext. Thanks to the introduction of tk in our definition, we can reach the property of being Fully Asymmetric. The proof of the following theorem is given in the full version of the paper.

Theorem 4. *The Anamorphic Encryption* NY *equppied with the Anamorphic Triplet* aNY *given in Fig. 5 is a Fully Asymmetric Anamorphic Encryption.*

$\mathsf{aGen}(\lambda)$

1 : $(\mathsf{pk}_0, \mathsf{sk}_0) \xleftarrow{\$} \Pi.\mathsf{KGen}(\lambda)$
2 : $(\mathsf{pk}_1, \mathsf{sk}_1) \xleftarrow{\$} \Pi.\mathsf{KGen}(\lambda)$
3 : $(\Sigma, \mathsf{aux}) \xleftarrow{\$} \Sigma.\mathsf{Sim}_0(\lambda)$
4 : $\mathsf{apk} = (\mathsf{pk}_0, \mathsf{pk}_1, \Sigma)$
5 : $\mathsf{ask} = \mathsf{sk}_0$
6 : $\mathsf{dk} = (\mathsf{pk}_0, \mathsf{pk}_1, \mathsf{aux})$
7 : $\mathsf{tk} = \mathsf{sk}_1$
8 : **return** $(\mathsf{apk}, \mathsf{ask}, \mathsf{dk}, \mathsf{tk})$

$\mathsf{aEnc}(\mathsf{apk}, \mathsf{dk}, m, \widehat{m})$

1 : $\mathsf{ct}_0 = \Pi.\mathsf{Enc}(\mathsf{pk}_0, m_0)$
2 : $\mathsf{ct}_1 = \Pi.\mathsf{Enc}(\mathsf{pk}_1, \widehat{m})$
3 : $\pi = \Sigma.\mathsf{Sim}_1((\mathsf{pk}_0, \mathsf{ct}_0), (\mathsf{pk}_1, \mathsf{ct}_1), \mathsf{aux})$
4 : $\mathsf{act} = (\mathsf{ct}_0, \mathsf{ct}_1, \pi)$
5 : **return** act

$\mathsf{aDec}(\mathsf{dk}, \mathsf{tk}, \mathsf{ask}, \mathsf{act})$

1 : Parse $\mathsf{act} = (\mathsf{ct}_0, \mathsf{ct}_1, \pi)$
2 : $\widehat{m} = \Pi.\mathsf{Dec}(\mathsf{sk}_1, \mathsf{ct}_1)$
3 : **return** $\widehat{m}$

Fig. 5. Anamorphic Triplet aNY.

Namely, for any PPT *distinguisher* $\mathcal{A}$ *that distinguish* $\mathsf{FAsyAnam\text{-}IND\text{-}CPA}^0_{\mathsf{NY},\mathsf{aNY}}$ *from* $\mathsf{FAsyAnam\text{-}IND\text{-}CPA}^1_{\mathsf{NY},\mathsf{aNY}}$ *there exists an adversary* $\mathcal{D}$ *such that*

$$\mathsf{Adv}^{\mathsf{FAsy\text{-}Anam}}_{\mathcal{A},\mathsf{aNY}}(\lambda) \leq 2 \cdot \mathsf{Adv}^{\text{IND-CPA}}_{\mathcal{D},\Pi}(\lambda)$$

Achieving Full Homomorphism. In [ADKL19] the first fully homomorphic NIZK construction for NP is given. Briefly, for a FH NIZK holds that evaluating on proofs that verify will result in a proof that verify and fresh proofs are indistinguishable from evaluated proofs.

Following the Naor-Yung transform paradigm, it is possibile to have a FH PKE scheme NY that is IND-CCA1 secure simply compiling a FH PKE scheme Π with a FH NIZK Σ.

The Eval algorithm of such scheme just takes ciphertexts as input and the function to apply to them and then use $\Pi.\mathsf{Eval}$ and $\Sigma.\mathsf{Eval}$ to obtain the new ciphertext.

Clearly, equipping this scheme with an Anamorphic Triplet aNY give us an Anamorphic Encryption scheme that has homomorphic properties, indeed it is a Fully Homomorphic Anamorphic Encryption scheme.

4.2 Cramer-Shoup Lite Gives Homomorphic Anamorphic Encryption

We show a concrete HAE construction based on the so called *Cramer-Shoup lite* (CS-lite for short) scheme [CS98], a well known, IND-CCA1 secure, variant of the *Cramer-Shoup* cryptosystem. We start by describing the basic scheme in Fig. 6.

Note that this is the *lifted* variant of the original scheme, (in [CS98] the message space is the cyclic group $\mathbb{G}$). In order to make decryption feasible, the message space is restricted to $\mathcal{M} = \{0, \ldots, B-1\}$, where $B = poly(\lambda)$.

$\mathsf{KGen}(\lambda)$

$\mathbb{G}, q \xleftarrow{\$} \mathcal{G}(\lambda)$
$g_1, g_2 \xleftarrow{\$} \mathbb{G}$
$x_1, x_2, z \xleftarrow{\$} \mathbb{Z}_q$
$c = g_1^{x_1} g_2^{x_2}$
$h = g_1^z$
$\mathsf{pk} = (g_1, g_2, c, h)$
$\mathsf{sk} = (x_1, x_2, z)$
return $(\mathsf{pk}, \mathsf{sk})$

$\mathsf{Enc}(\mathsf{pk}, m)$

$r \xleftarrow{\$} \mathbb{Z}_q$
$u_1 = g_1^r$
$u_2 = g_2^r$
$e = h^r g_1^m$
$v = c^r$
$\mathsf{ct} = (u_1, u_2, e, v)$
return ct

$\mathsf{Dec}(\mathsf{sk}, \mathsf{ct})$

$m = \perp$
if $v = u_1^{x_1} u_2^{x_2}$ **then**
  $d = e/u_1^z$
  for $i \in \{0, \ldots, B-1\}$
    if $g_1^i = d$ **then**
      $m = i$
return m

Fig. 6. CS-lite encryption scheme.

CS-lite scheme is also a linearly homomorphic scheme. We next give two algorithms $\mathsf{EvalScal}$ and $\mathsf{EvalSum}$. $\mathsf{ct}, \mathsf{ct}_1$ and ct_2 are elements of the ciphertexts space, while α is a constant in the message space.

$\mathsf{EvalScal}(\mathsf{pk}, \mathsf{ct}, \alpha)$

Parse ct as (u_1, u_2, e, v)
$r' \xleftarrow{\$} \mathbb{Z}_q$
$u_1' = u_1^\alpha g_1^{r'}$
$u_2' = u_2^\alpha g_2^{r'}$
$e' = e^\alpha h^{r'}$
$v' = v^\alpha c^{r'}$
return (u_1', u_2', e', v')

$\mathsf{EvalSum}(\mathsf{pk}, \mathsf{ct}_1, \mathsf{ct}_2)$

Parse ct_1 as (u_1, u_2, e, v)
Parse ct_2 as (u_1, u_2, e, v)
$r' \xleftarrow{\$} \mathbb{Z}_q$
$u_1' = \mathsf{ct}_1.u_1 \cdot \mathsf{ct}_2.u_1 \cdot g_1^{r'}$
$u_2' = \mathsf{ct}_1.u_2 \cdot \mathsf{ct}_2.u_2 \cdot g_2^{r'}$
$e' = \mathsf{ct}_1.e \cdot \mathsf{ct}_2.e \cdot h^{r'}$
$v' = \mathsf{ct}_1.v \cdot \mathsf{ct}_2.v \cdot c^{r'}$
return (u_1', u_2', e', v')

Since the message space is restricted to $\{0, \ldots, B-1\}$ the number of possible homomorphic operations is limited so that the result of the final operation is less than B.

Anamorphic Construction. In this case we don't provide an anamorphic extension but rather an anamorphic triplet. The reason is that, to decrypt anamorphic ciphertexts, the scheme relies on a trapdoor tk that has to be created at key generation time. This trapdoor will be used by the receiver Bob to decrypt the anamorphic ciphertexts and, as already discussed before, is kept separate with respect to the double key dk (shared with Alice) as it is not needed to produce (anamorphic) ciphertexts. As we will prove below, keeping tk and dk separate is what allows us to achieve the property of being Fully Asymmetric. The anamorphic triplet $\mathsf{aCS} = (\mathsf{aGen}, \mathsf{aEnc}, \mathsf{aDec})$ is specified in Fig. 7.

Homomorphic Properties. Addition of plaintexts (both regular and anamorphic ones) is done using $\mathsf{EvalSum}$ as in regular CS-lite. Indeed, let act_1 and act_2 be the anamorphic ciphertexts corresponding to $(m_1, \widehat{m}_1), (m_2, \widehat{m}_2)$. Then

$\mathsf{aGen}(\lambda)$

$\mathbb{G}, q \xleftarrow{\$} \mathcal{G}$
$g_1 \xleftarrow{\$} \mathbb{G}$
$x \xleftarrow{\$} \mathbb{Z}_q$
$g_2 = g_1^x$
$s \xleftarrow{\$} \mathbb{Z}_q$
$x_1, x_2, z \xleftarrow{\$} \mathbb{Z}_q$
$c = g_1^{x_1} g_2^{x_2}$
$h = g_1^z$
$\mathsf{pk} = (g_1, g_2, c, h)$
$\mathsf{sk} = (x_1, x_2, z)$
$\mathsf{ppk} = (g_1^s, g_2^s g_1, h^s, c^s g_1^{x_2})$
$\mathsf{apk} = \mathsf{pk}$
$\mathsf{ask} = \mathsf{sk}$
$\mathsf{tk} = x$
$\mathsf{dk} = (\mathsf{pk}, \mathsf{ppk})$
return $(\mathsf{apk}, \mathsf{ask}, \mathsf{dk}, \mathsf{tk})$

$\mathsf{aEnc}(\mathsf{apk}, \mathsf{dk}, m, \widehat{m})$

Parse $\mathsf{ppk} = (\mathsf{up}_1, \mathsf{up}_2, \mathsf{hp}, \mathsf{cp})$
$r \xleftarrow{\$} \mathbb{Z}_q$
$u_1 = \mathsf{up}_1^{\widehat{m}} g_1^r$
$u_2 = \mathsf{up}_2^{\widehat{m}} g_2^r$
$e = \mathsf{hp}^{\widehat{m}} h^r g_1^m$
$v = \mathsf{cp}^{\widehat{m}} c^r$
$\mathsf{act} = (u_1, u_2, e, v)$
return act

$\mathsf{aDec}(\mathsf{dk}, \mathsf{tk}, \mathsf{ask}, \mathsf{act})$

Parse $\mathsf{act} = (u_1, u_2, e, v)$
$d = u_2 / u_1^x$
for $i \in \{0, \ldots, B-1\}$
  if $g_1^i = d$ **then**
    return i
return $\perp$

Fig. 7. Anamorphic Triplet aCS.

$$\begin{aligned}\mathsf{EvalSum}(\mathsf{apk}, \mathsf{act}_1, \mathsf{act}_2) = (&u_1' = g_1^{s\widehat{m}_1} g_1^{r_1} g_1^{s\widehat{m}_2} g_1^{r_2} g_1^{r'},\\ &u_2' = g_2^{s\widehat{m}_1} g_1^{\widehat{m}_1} g_2^{r_1} g_2^{s\widehat{m}_2} g_1^{\widehat{m}_2} g_2^{r_2} g_2^{r'},\\ &e' = h^{s\widehat{m}_1} h^{r_1} g_1^{m_1} h^{s\widehat{m}_2} h^{r_2} g_1^{m_2} h^{r'},\\ &v' = c^{s\widehat{m}_1} g_1^{x_2\widehat{m}_1} c^{r_1} c^{s\widehat{m}_2} g_1^{x_2\widehat{m}_2} c^{r_2} c^{r'})\end{aligned}$$

setting $t = r_1 + r_2 + r', m' = m_1 + m_2, \widehat{m}' = \widehat{m}_1 + \widehat{m}_2$, this becomes:

$$\begin{aligned}&\mathsf{EvalSum}(\mathsf{apk}, \mathsf{act}_1, \mathsf{act}_2) =\\ &\quad = (u_1' = \mathsf{up}_1^{\widehat{m}'} g_1^t, u_2' = \mathsf{up}_2^{\widehat{m}'} g_2^t, e' = \mathsf{hp}^{\widehat{m}'} h^t g_1^{m'}, v' = \mathsf{cp}^{\widehat{m}'} c^t)\end{aligned}$$

which is distributed as a fresh output of $\mathsf{aEnc}(\mathsf{apk}, \mathsf{dk}, m_1 + m_2, \widehat{m}_1 + \widehat{m}_2)$.

Similarly, multiplication by a scalar α is done using EvalScal as in the base scheme. Let act be the anamorphic ciphertext corresponding to $(m, \widehat{m})$. Then

$$\begin{aligned}\mathsf{EvalScal}(\mathsf{apk}, \mathsf{act}, \alpha) = ((g_1^{s\widehat{m}}(g_1^r))^\alpha g_1^{r'}, &(g_2^{s\widehat{m}} g_1^{\widehat{m}} g_2^r)^\alpha g_2^{r'},\\ &(h^{s\widehat{m}} h^r g_1^m)^\alpha h^{r'}, (c^{s\widehat{m}} g_1^{x_2\widehat{m}} c^r)^\alpha c^{r'})\end{aligned}$$

We can rewrite the previous equation setting $t = \alpha r + r', m' = \alpha \cdot m, \widehat{m}' = \alpha \cdot \widehat{m}$:

$$\mathsf{EvalScal}(\mathsf{apk}, \mathsf{act}, \alpha) = (u_1' = \mathsf{up}_1^{\widehat{m}'} g_1^t, u_2' = \mathsf{up}_2^{\widehat{m}'} g_2^t, e' = \mathsf{hp}^{\widehat{m}'} h^t g_1^{m'}, v' = \mathsf{cp}^{\widehat{m}'} c^t)$$

which is distributed as expected.

In the next theorem (whose proof appears the full version of the paper) we prove that scheme is strongly homomorphic.

Theorem 5. *The anamorphic triplet for lifted Cramer-Shoup lite* aCS *given in Fig. 7 is perfectly strongly homomorphic for the class of linear functions.*

Anamorphism. In the following theorem we prove that the scheme is anamorphic according to our definition from Sect. 2.5, the proof appears in the full version of the paper.

Theorem 6. *If DDH holds then Cramer-Shoup lite cryptosystem equipped with the anamorphic triplet* aCS *given in Fig. 7 is an anamorphic encryption scheme. Namely, for any* PPT *distinguisher* $\mathcal{D}$ *that distinguishes* $\mathsf{RealG}_{\mathsf{CS}}$ *from* $\mathsf{AnamorphicG}_{\mathsf{aCS}}$ *there exists an adversary* $\mathcal{B}$ *such that*

$$\mathsf{Adv}^{\mathsf{Anamorphism}}_{\mathcal{D},\mathsf{CS},\mathsf{aCS}}(\lambda) \leq 2 \cdot \mathsf{Adv}^{\mathrm{DDH}}_{\mathcal{B}}(\lambda)$$

Fully Asymmetric. In the following theorem we show that the scheme also satisfies the property of being Fully Asymmetric Sect. 2.6, the proof appears in the full version of the paper.

Theorem 7. *If DDH holds then the Anamorphic Encryption* CS *equppied with the Anamorphic Triplet* aCS *given in Fig. 7 is a Fully Asymmetric Anamorphic Encryption. Namely, for any* PPT *distinguisher* $\mathcal{D}$ *that distinguishes games* $\mathsf{FAsyAnam\text{-}IND\text{-}CPA}^0_{\mathsf{aCS}}$ *and* $\mathsf{FAsyAnam\text{-}IND\text{-}CPA}^1_{\mathsf{aCS}}$ *there exists an adversary* $\mathcal{B}$ *such that*

$$\mathsf{Adv}^{\mathsf{FAsy\text{-}Anam}}_{\mathcal{D},\mathsf{aCS}}(\lambda) \leq 4 \cdot \mathsf{Adv}^{\mathrm{DDH}}_{\mathcal{B}}(\lambda)$$

4.3 GSW Gives Homomorphic Anamoprhic Encryption

GSW Notation and Construction. In this section we show that the fully homomorphic encryption proposed by Gentry, Sahai and Waters [GSW13] can be turned into an anamorphic scheme retaining the homomorphic properties.

First let us recover some notation from the original paper: $\mathbf{2}^\ell$ is the vector of powers of two $(1, 2, \ldots, 2^{\ell-1})$. G_r^{-1} is the *bit decomposition* operations, i.e. $G_r^{-1}(x) = (x_0, \ldots, x_{\ell-1})$ such that $x = x_0 + \ldots 2^{\ell-1} x_{\ell-1}$. This is extended to vectors by concatenating all decompositions and to matrices by applying it row-wise. Thus for any $A \in \mathbb{Z}_q^{n,m}$, we have $G_r^{-1}(A) \in \mathbb{Z}_q^{n,\ell m}$. G_r is the inverse operation, such that $G_r(x_0, \ldots, x_{\ell-1}) = x_0 + \ldots + 2^{\ell-1} x_{\ell-1}$. As before this is extended to matrices acting row-wise. $\otimes$ is the Kronecker product, such that $\mathbf{a} \otimes \mathbf{b} = (a_1 \mathbf{b}, \ldots, a_n \mathbf{b})$. We write $(\mathbf{v}, M)$ to append the vector $\mathbf{v}$ to the matrix M column-wise.

Given the above definitions we recall three main properties from [GSW13]:

Proposition 1. *For any* $A \in \mathbb{Z}_q^{m,n\ell}$, $\mathbf{b} \in \mathbb{Z}_q^n$, $C \in \mathbb{Z}_q^{m,n}$ *then:*

1. G_r *is a linear map, i.e.* $G_r(A_1 + A_2) = G_r(A_1) + G_r(A_2)$.
2. $G_r^{-1}(A) \cdot (\mathbf{b} \otimes \mathbf{2}^\ell) = A\mathbf{b}$
3. $C \cdot (\mathbf{b} \otimes \mathbf{2}^\ell) = G_r(C) \cdot \mathbf{b} = G_r^{-1}(G_r(C)) \cdot (\mathbf{b} \otimes \mathbf{2}^\ell)$.

We are now ready to recall the GSW encryption scheme, with a full description appearing in Fig. 8. Regarding the parameters used, n, m, q, χ are chosen so that $\mathsf{LWE}_{m,n,q,\chi}$ is hard, with n being the lattice dimension, m the number of LWE samples, q the modulus and χ the error distribution. To guarantee security [GSW13] further requires $m \geq 2n \log_2 q$. Finally, $\ell := \lfloor \log_2 q \rfloor + 1$ and $N := n \cdot \ell$.

$\mathsf{KGen}(\lambda)$

1 : $B \xleftarrow{\$} \mathbb{Z}_q^{m,n-1}$, $\mathbf{t} \xleftarrow{\$} \mathbb{Z}_q^{n-1}$, $\mathbf{e} \xleftarrow{\$} \chi^m$
2 : $\mathbf{b} = B\mathbf{t} + \mathbf{e}$ // LWE sample
3 : $A = (\mathbf{b}, B)$
4 : $\mathbf{s} = (1, -\mathbf{t})$
5 : $\mathbf{v} = \mathbf{s} \otimes \mathbf{2}^\ell$
6 : **return** $(\mathsf{pk}, \mathsf{sk}) = (A, \mathbf{v})$

$\mathsf{Enc}(\mathsf{pk}, \mu)$

1 : $R \xleftarrow{\$} \{0,1\}^{N,m}$
2 : $C = G_r^{-1} \circ G_r \left(\mu \cdot I_N + G_r^{-1}(RA)\right)$
3 : **return** C

$\mathsf{Dec}(\mathsf{sk}, C)$

1 : **if** $C\mathbf{v} = \mu\mathbf{v} + \mathbf{e}'$ with suitably short $\mathbf{e}'$:
2 : Extract μ as in [GSW13, MP12]
3 : **return** μ

Fig. 8. Original GSW fully homomorphic encryption scheme.

Anamorphism. We now present our anamorphic version of GSW as described in Fig. 8. The main idea in the original scheme is to encrypt μ as the eigenvalue of a secret approximate eigenvector $\mathbf{v}$. In our anamorphic construction we modify the public parameters generation so that a ciphertext C can be created with two secret approximate eigenvectors $\mathbf{v}_1, \mathbf{v}_2$. Specifically C will satisfy $C\mathbf{v}_1 = \mu_1\mathbf{v}_1 + \mathbf{e}'_1$ with μ_1 being the regular message, whereas $C\mathbf{v}_2 = \mu_2\mathbf{v}_2 + \mathbf{e}'_2$ with μ_2 being the anamorphic message. A full description of the scheme is presented in Fig. 9.

First we observe that even in anamorphic mode the scheme remains homomorphic using the same argument from the original paper. Indeed, given $C, \widehat{C}$ encrypting in anamorphic mode $\mu, \widehat{\mu}$ then by correctness $C\mathbf{v}_2 = \mu\mathbf{v}_2 + \mathbf{e}$ and $\widehat{C}\mathbf{v}_2 = \widehat{\mu}\mathbf{v}_2 + \widehat{\mathbf{e}}$. Thus $C + \widehat{C}$ is an encryption of $\mu + \widehat{\mu}$ and the product $C \cdot \widehat{C}$ encrypts $\mu \cdot \widehat{\mu}$ assuming that the resulting errors, respectively $\mathbf{e} + \widehat{\mathbf{e}}$ and $\widehat{\mu}\mathbf{e} + C\widehat{\mathbf{e}}$, have low norm.

Note that the matrices P_1, P_2 described in line 2 can be computed in any modulus q (not necessarily prime) and without knowing $\mathbf{v}_2$. An examples of such pair can be based on the fact that by construction $\mathbf{v}_1 = (1, \widetilde{\mathbf{v}}_1)$ and $\mathbf{v}_2 = (0, \widetilde{\mathbf{v}}_2)$:

$\mathsf{aGen}(\lambda)$

$\widetilde{B} \xleftarrow{\$} \mathbb{Z}_q^{m,n-2}$
$\mathbf{t}_1 \xleftarrow{\$} \mathbb{Z}_q^{n-1}$, $\mathbf{t}_2 \xleftarrow{\$} \mathbb{Z}_q^{n-2}$
Sample errors $\mathbf{e}_1, \mathbf{e}_2 \xleftarrow{\$} \chi^m$
$\mathbf{b}_2 = \widetilde{B}\mathbf{t}_2 + \mathbf{e}_2$ and $B = (\mathbf{b}_2, \widetilde{B})$
$\mathbf{b}_1 = B\mathbf{t}_1 + \mathbf{e}_1$
$A = (\mathbf{b}_1, B)$
$\mathbf{s}_1 = (1, -\mathbf{t}_1)$
$\mathbf{s}_2 = (0, 1, -\mathbf{t}_2)$
$\mathbf{v}_1 = \mathbf{s}_1 \otimes \mathbf{2}^\ell$
$\mathbf{v}_2 = \mathbf{s}_2 \otimes \mathbf{2}^\ell$
$\mathsf{apk} = A,\ \mathsf{ask} = \mathbf{v}_1$
$\mathsf{tk} = \mathbf{v}_2,\ \mathsf{dk} = \mathbf{v}_1$
return $(\mathsf{apk}, \mathsf{ask}, \mathsf{tk}, \mathsf{dk})$

$\mathsf{aEnc}(\mathsf{apk}, \mathsf{dk}, \mu_1, \mu_2)$

$R \xleftarrow{\$} \{0,1\}^{N,m}$
Compute $P_1, P_2 \in \mathbb{Z}_q^{N,N}$ such that:
  $P_i\mathbf{v}_i = \mathbf{v}_i$
  $P_i\mathbf{v}_j = \mathbf{0}$ for $i \neq j$
$C' = \mu_1 P_1 + \mu_2 P_2 + G_r^{-1}(RA)$
$C = G_r^{-1} \circ G_r\,(C')$
return C

$\mathsf{aDec}(\mathsf{dk}, \mathsf{tk}, C)$

if $C\mathbf{v}_2 = \mu_2\mathbf{v}_2 + \mathbf{e}'$ with suitably short $\mathbf{e}'$:
  Extract μ_2 as in [GSW13, MP12]
  return μ_2

Fig. 9. Anamorphic Triplet for the GSW scheme.

$$P_1 = (\mathbf{v}_1, \Omega_{N,N-1}) \quad \Rightarrow \quad \begin{cases} P_1\mathbf{v}_1 = 1 \cdot \mathbf{v}_1 + \Omega_{N,N-1}\widetilde{\mathbf{v}}_1 = \mathbf{v}_1 \\ P_1\mathbf{v}_2 = 0 \cdot \mathbf{v}_2 + \Omega_{N,N-1}\widetilde{\mathbf{v}}_2 = \mathbf{0}. \end{cases}$$

where $\Omega_{n,m} \in \mathbb{Z}_q^{n,m}$ is the zero matrix. Given P_1 we can then set $P_2 = I_N - P_1$.

We finally remark that our proof for Theorem 8 is tight, i.e. our bound on the adversary's advantage does not depend on the number of encryption queries.

Theorem 8. *The GSW scheme described in Fig. 8 is an anamorphic encryption scheme, with anamorphic triplet* $(\mathsf{aGen}, \mathsf{aEnc}, \mathsf{aDec})$ *as defined in Fig. 9, assuming* $\mathsf{LWE}_{m,n-2,q,\chi}$ *and parameters satisfying* $m \geq n \log_2 q + 2\lambda/N$.

Proof. We will prove that the triple $(\mathsf{aGen}, \mathsf{aEnc}, \mathsf{aDec})$ is anamorphic through a sequence of hybrid games:

- G_0: The standard Anamorphic Game.
- G_1: As G_0, but the parameters $(\mathsf{apk}, \mathsf{ask}, \mathsf{tk}, \mathsf{dk})$ are computed through the hybrid KGen_1 defined in Fig. 10.
- G_2: As G_1, but when $\mathcal{A}$ requests an encryption of (μ_1, μ_2), the ciphertext is computed as $C = \mathsf{Enc}_2(\mathsf{apk}, \mathsf{dk}, \mu_1, \mu_2)$, see Fig. 10.
- G_3: As G_1, but when $\mathcal{A}$ requests an encryption of (μ_1, μ_2), the ciphertext is computed as $C = \mathsf{Enc}_3(\mathsf{apk}, \mathsf{dk}, \mu_1, \mu_2)$, see Fig. 10.
- G_4: The Real Game, where the keys are generated with $\mathsf{pk}, \mathsf{sk} \xleftarrow{\$} \mathsf{KGen}(\lambda)$ and the challenge ciphertext is $\mathsf{Enc}(\mathsf{pk}, \mu_1)$.

$\mathsf{KGen}_1(\lambda)$

$\mathbf{t}_1 \xleftarrow{\$} \mathbb{Z}_q^{n-1}, \mathbf{t}_2 \xleftarrow{\$} \mathbb{Z}_q^{n-2}$

$\mathbf{e}_1, \mathbf{e}_2 \xleftarrow{\$} \chi^m$

$B \xleftarrow{\$} \mathbb{Z}_q^{m,n-1}$

$\mathbf{b}_1 = B\mathbf{t}_1 + \mathbf{e}_1$

$A = (\mathbf{b}_1, B)$

$\mathbf{s}_1 = (1, -\mathbf{t}_1)$, $\mathbf{s}_2 = (0, 1, -\mathbf{t}_2)$

$\mathbf{v}_1 = \mathbf{s}_1 \otimes \mathbf{2}^\ell$, $\mathbf{v}_2 = \mathbf{s}_2 \otimes \mathbf{2}^\ell$

$\mathsf{apk} = A$, $\mathsf{ask} = \mathbf{v}_1$

$\mathsf{tk} = \mathbf{v}_2$, $\mathsf{dk} = \mathbf{v}_1$

return $(\mathsf{apk}, \mathsf{ask}, \mathsf{tk}, \mathsf{dk})$

$\mathsf{Enc}_2(\mathsf{apk}, \mathsf{dk}, \mu_1, \mu_2)$

Sample $R \xleftarrow{\$} \{0,1\}^{N,m}$

Compute P_1, P_2 as in aEnc given $\mathbf{v}_1$

Sample $S \xleftarrow{\$} \mathbb{Z}_q^{N,n-1}$

Compute $\bar{R} = (R\mathbf{e}_1 + S\mathbf{t}_1, S)$

$C = G_r^{-1} \circ G_r \left(\mu_1 P_1 + \mu_2 P_2 + G_r^{-1}(\bar{R})\right)$

return C

$\mathsf{Enc}_3(\mathsf{apk}, \mathsf{dk}, \mu_1, \mu_2)$

Sample $R \xleftarrow{\$} \{0,1\}^{N,m}$

Compute P_1, P_2 as in aEnc given $\mathbf{v}_1$

Sample $S \xleftarrow{\$} \mathbb{Z}_q^{N,n-1}$

Compute $\bar{R} = (R\mathbf{e}_1 + S\mathbf{t}_1, S)$

$C = G_r^{-1} \circ G_r \left(\mu_1 I_N + G_r^{-1}(\bar{R})\right)$

return C

Fig. 10. Hybrid Key Generation and Encryption for the proof of Theorem 8. Introduced differences are highlighted.

At a high level we will reduce the indistinguishability of $\mathsf{G}_0, \mathsf{G}_1$ to an LWE instance. Next we show $\mathsf{G}_1, \mathsf{G}_2$ and $\mathsf{G}_3, \mathsf{G}_4$ are statistically close using the Leftover Hash Lemma, and finally observe G_2, G_3 produces identical distributions. Within the scope of this proof $x \sim U(X)$ means that x is a random variable uniformly distributed over a finite set X.

Proof of $\mathsf{G}_0 \overset{c}{=} \mathsf{G}_1$. For any distinguisher $\mathcal{D}$ we describe an adversary $\mathcal{B}$ breaking $\mathsf{LWE}_{m,n-2,q,\chi}$. The idea is to simply use the LWE samples as the matrix $\widetilde{B}$ and vector $\mathbf{b}_2$ in the parameter generation. Remarkably, although $\mathcal{B}$ will be unable to compute $\mathbf{v}_2$, this value is unnecessary to produce the challenge ciphertext or the keys observed by $\mathcal{D}$, namely apk, ask. A full description of $\mathcal{B}$ appears in Fig. 11 for completeness.

By inspection it is easy to observe that when $\mathbf{b}^*$ is randomly sampled, then $B \sim U(\mathbb{Z}_q^{m,n})$ and in particular $\mathcal{B}$ perfectly simulates G_1. Conversely, if $\mathbf{b}^* = A^*\mathbf{t} + \mathbf{e}$ with $\mathbf{t} \sim U(\mathbb{Z}_q^{n-2})$ and $\mathbf{e} \sim \chi^m$, then $(A, \mathbf{v}_1, C)$ are distributed as in G_0. We thus conclude that $\mathsf{Adv}_{\mathcal{D}}^{\mathsf{G}_0,\mathsf{G}_1}(\lambda) = \mathsf{Adv}_{\mathcal{B}}^{\mathrm{LWE}}(\lambda)$ which is negligible if $\mathsf{LWE}_{m,n-2,q,\chi}$ is hard.

Proof of $\mathsf{G}_1 \overset{s}{=} \mathsf{G}_2$. This part is based on the Leftover Hash Lemma which we restate here. Notation-wise $\Delta(x, y)$ is the statistical distance of two random variables x, y; $\mathrm{H}_\infty(x)$ is the min-entropy of x, and $\mathrm{H}_\infty(x|y)$ the conditional min-entropy of x with respect to y.

$\mathcal{B}(A^*, \mathbf{b}^*)$

1 : // Note $A^* \in \mathbb{Z}_q^{m,n-2}$ and $\mathbf{b}^* \in \mathbb{Z}_q^m$
2 : Set $B = (\mathbf{b}^*, A^*)$ as the column-wise concatenation
3 : Sample $\mathbf{t}_1 \xleftarrow{\$} \mathbb{Z}_q^{n-2}$ and $\mathbf{e}_1 \xleftarrow{\$} \chi^m$
4 : Set $\mathbf{b}_1 = B\mathbf{t}_1 + \mathbf{e}_1$ and $A = (\mathbf{b}_1, B)$
5 : $\mathbf{s}_1 = (1, -\mathbf{t}_1)$ and $\mathbf{v}_1 = \mathbf{s}_1 \otimes \mathbf{2}^\ell$
6 : Set $\mathsf{apk} = A$, $\mathsf{ask} = \mathbf{v}_1$ and execute $\mathcal{D}(\mathsf{apk}, \mathsf{ask})$
7 : **when** $\mathcal{D}$ queries an encryption of (μ_1, μ_2):
8 : Compute P_1, P_2 as in aEnc using $\mathbf{v}_1$
9 : Sample $R \xleftarrow{\$} \{0,1\}^{N,m}$
10 : $C = G_r^{-1} \circ G_r \left(\mu_1 P_1 + \mu_2 P_2 + G_r^{-1}(RA)\right)$
11 : Send the challenge ciphertext C to $\mathcal{D}$
12 : **when** $\mathcal{D} \to b$: **return** b

Fig. 11. $\mathcal{B}$ reducing a distinguisher $\mathcal{D}$ for $\mathsf{G}_0, \mathsf{G}_1$ to $\mathsf{LWE}_{m,n-2,q,\chi}$.

Lemma 2 (Leftover Hash Lemma [ILL89]). *Let $x \sim \mathcal{X}$, z be random variables and $\mathcal{H}$ be a family of universal hash function with domain $\mathcal{X}$ and image $\mathcal{Y}$. Sampling $y \sim U(\mathcal{Y})$ and $h \sim U(\mathcal{H})$, if $k = \mathrm{H}_\infty(x|z)$ and $m = \log_2(|\mathcal{Y}|)$ and $y \sim U(\mathcal{Y})$, then*

$$m \leq k - 2\log_2(1/\varepsilon) \quad \Rightarrow \quad \Delta\left((h, h(x), z),\, (h, y, z)\right) \;\leq\; \varepsilon/2.$$

Let p be an upper bound on the number of queries a distinguisher between $\mathsf{G}_1, \mathsf{G}_2$ would make. Then we will use this bound to show that, using the same notation as game G_1 and G_2

$$\Delta\left((B, (R_i B, R_i \mathbf{e}_1)_{i=1}^p),\; (B, (S_i, R_i \mathbf{e}_1)_{i=1}^p)\right) \leq 2^{-(\lambda-1)}.$$

where R_i and S_i are the random matrices sampled to compute the i-th challenge ciphertexts in the two games. In this direction we first point out that $\mathbb{Z}_q^{m,n-1}$ is an almost universal hash function family from $\{0,1\}^{N,m}$ to $\mathbb{Z}_q^{N,n-1}$ for any modulus q. In particular, the entry-wise application of B to a vector of matrices in $(\{0,1\}^{N,m})^p$ it also a universal hash to $(\mathbb{Z}_q^{N,n-1})^p$. Next we bound the conditional min-entropy of R_i given $R_i\mathbf{e}_1$:

$$\begin{aligned}
\mathrm{H}_\infty(R_i|R_i\mathbf{e}_1) \;&\geq\; \mathrm{H}_\infty(R_i) - \mathrm{H}_\infty(R_i\mathbf{e}_1) \\
&\geq\; \mathrm{H}_\infty(R_i) - \log_2|\mathbb{Z}_q^N| \\
&\geq\; Nm - N\log_2 q \;=\; N(m - \log_2 q).
\end{aligned}$$

Because all R_i are sampled independently we can then bound the conditional min-entropy of $(R_1, \ldots, R_p)$ given $(R_1\mathbf{e}_1, \ldots, R_p\mathbf{e}_1)$ as

$$\mathrm{H}_\infty((R_i)_{i=1}^p|(R_i\mathbf{e}_1)_{i=1}^p) \;=\; \sum\nolimits_{i=1}^n \mathrm{H}_\infty(R_i|R_i\mathbf{e}_i) \;\leq\; pN(m - \log_2 q).$$

The leftover hash lemma can then be applied because

$$\begin{aligned}\log_2 \left|(\mathbb{Z}_q^{N,n-1})^p\right| \;&=\; pN(n-1)\log_2 q \;\leq \\ &\leq\; pN(m-\log_2 q) - 2\lambda \;\leq\; \mathrm{H}_\infty((R_i)_{i=1}^p | (R_i\mathbf{e}_1)_{i=1}^p) - 2\lambda\end{aligned}$$

where the first inequality follows as we assumed $m \geq n\log_2 q + 2\lambda/N$. Now that we proved the above statistical distance to be small, it is easy to observe that

$$\Delta\left((B, \mathbf{t}_1, \mathbf{e}_1, (R_iB, R_i\mathbf{e}_1)_{i=1}^p),\, (B, \mathbf{t}_1, \mathbf{e}_1, (S_i, R_i\mathbf{e}_1)_{i=1}^p)\right) \;\leq\; 2^{-(\lambda-1)}$$

as $\mathbf{t}_1$ is independent from the other variables and $\mathbf{e}_1$ conditioned on $R_i\mathbf{e}_1$ follows the same distribution in both vectors. Finally, as the messages produced in G_1, G_2 are deterministic functions of these random variables, we conclude the two games to be statistically close.

Proof of $\mathsf{G}_2 \stackrel{p}{=} \mathsf{G}_3$. To show that the two distributions are the same, we observe that one can be obtained from the other up to applying a linear (bijective) map on S. We begin by observing that the matrix $G_r(\mu_1 P_1 + \mu_2 P_2 - \mu_1 I_N)$ is of the form $(S_0\mathbf{t}_1, S_0)$ for some $S_0 \in \mathbf{t}$. This hold because

$$\begin{aligned}&(\mu_1 P_1 + \mu_2 P_2 - \mu_1 I_N)\mathbf{v}_1 \;=\; \mu_1\mathbf{v}_1 + \mathbf{0} - \mu_1\mathbf{v}_1 \;=\; \mathbf{0}. \\ \Rightarrow\quad &G_r(\mu_1 P_1 + \mu_2 P_2 - \mu_1 I_N)\mathbf{s}_1 \;=\; (\mu_1 P_1 + \mu_2 P_2 - \mu_1 I_N)\cdot(\mathbf{s}_1 \otimes \mathbf{2}^\ell) \;=\; \mathbf{0}\end{aligned}$$

where the third equality uses Proposition 1. Moreover any matrix M such that $M\mathbf{s}_1 = \mathbf{0}$ is of the desired form because, calling $M = (\mathbf{u}, S_0)$

$$M\mathbf{s}_1 = \mathbf{0} \quad\Rightarrow\quad \mathbf{0} \;=\; (\mathbf{u}, S_0)(1, -\mathbf{t}_1) \;=\; \mathbf{u} - S_0\mathbf{t}_1 \quad\Rightarrow\quad \mathbf{u} = S_0\mathbf{t}_1.$$

To conclude we show that replacing $S \mapsto S - S_0$ in the encryption algorithm in G_2, produces the distribution in G_3.

$$\begin{aligned}C \;&=\; G_r^{-1} \circ G_r\left(\mu_1 P_1 + \mu_2 P_2 + G_r^{-1}(\bar{R} - (S_0\mathbf{t}_1, S_0))\right) \\ &=\; G_r^{-1}\left(G_r\left(\mu_1 P_1 + \mu_2 P_2\right) + \bar{R} - (S_0\mathbf{t}_1, S_0)\right) \\ &=\; G_r^{-1}\left(G_r\left(\mu_1 I_N\right) + G_r \circ G_r^{-1}(\bar{R})\right) \\ &=\; G_r^{-1} \circ G_r\left(\mu_1 I_N + G_r(\bar{R})\right)\end{aligned}$$

where the first equality follows by the linearity of G_r, the second again by the linearity of G_r and since $(S_0\mathbf{t}_1, S_0) = G_r(\mu_1 P_1 + \mu_2 P_2) - G_r(\mu_1 I_N)$. The third equality instead uses the fact that $G_r \circ G_r^{-1}$ is the identity function. As the distribution of S and $S - S_0$ is identical for $S \sim U(\mathbb{Z}_q^{N,n-1})$ we conclude that G_2, G_3 follow the same distribution.

Proof of $\mathsf{G}_3 \stackrel{s}{=} \mathsf{G}_4$. The proof is identical to the one for the case $\mathsf{G}_1 \stackrel{s}{=} \mathsf{G}_2$. Using the same notation for p, R_i and S_i, we can argue with the Leftover Hash Lemma that the two distributions

$$\left(B, (R_iB, R_i\mathbf{e}_1)_{i=1}^p\right), \quad \left(B, (S_i, R_i\mathbf{e}_1)_{i=1}^p\right)$$

are statistically close. This further implies that the random variables

$$(B, \mathbf{t}_1, \mathbf{e}_1, (R_i B, R_i \mathbf{e}_1)_{i=1}^p), \quad (B, \mathbf{t}_1, \mathbf{e}_1, (S_i, R_i \mathbf{e}_1)_{i=1}^p)$$

are statistically close. Finally, in G_3 the keys sent to the adversary are $\mathsf{apk} = A = (\mathbf{b}_1, B)$ with $B \sim U(\mathbb{Z}_q^{m,n-2})$ and $\mathbf{b} = B\mathbf{t}_1 + \mathbf{e}_1$; $\mathsf{ask} = (1, -\mathbf{t}_1)$. Thus the views in the two games are obtained applying the same deterministic function on the two vectors above. We conclude G_3 and G_4 are statistically indistinguishable.

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